

Assignment 1



Q1)

→ <u>Conventional Energy</u>	<u>Non-Conventional Energy</u>
1) Traditional Sources used for a long time	Newer and alternative sources developed recently
2) Coal, Petroleum, Natural Gas, Hydropower	Solar, Wind, Biomass, Tidal, Geothermal
3) Limited and exhaustible	Abundant and naturally replenished.
4) High pollution (CO_2 , smoke)	Low or almost no pollution.
5) Lower initial cost but high fuel cost. Not-	Higher initial cost but low operating cost
6) Sustainable for future	Sustainable & eco-friendly

Q2)

→ <u>Renewable Energy</u>	<u>Non-Renewable Energy</u>
1) Energy sources that can be replenished naturally	Energy sources that get depleted once used.
2) Solar, Wind, Hydro, Biomass, Geothermal	Coal, Oil, Natural gas, Nuclear fuels.
3) Unlimited	Limited and will finish over time.
4) Clean energy & less harmful	Cause pollution (greenhouse gases)
5) High setup cost but free fuel.	Lower setup but expensive fuel.

Q3

→ Solar PV SystemSolar Thermal System

1) Converts light into electricity

Converts sunlight into heat energy

2) Electrical energy (DC)

Thermal energy (hot water)

3) Solar PV cells

Solar collectors, heat exchanger

4)

Lighting, solar pumps, rooftop power generation

Water heating, industrial heating, steam power plants

5) Moderate efficiency
15% - 25%Higher for heating
(20% - 70%)

6) Low maintainance

Moderate maintenance

7) Higher cost per unit

Less effective for heating application